

SOFTWARE REQUIREMENTS SPECIFICATION

Components Inventory Management System (software only)

Electronic component catalog, physical drawer mapping, stock control and requisitions

1. Functional Requirements

1.1 Component Library

FR-1.1 A component record shall hold the following fields.

Field	Type	Required	Notes
Image	File	No	jpg, png or webp, max 2 MB, auto thumbnailed
Name	Text	Yes	Indexed for search
Part number / MPN	Text	No	Unique when supplied
Category	Reference	Yes	Single category per component
Unit	Reference	Yes	Determines whether fractional quantities are allowed
Description	Long text	No	Rich text
Last buying price	Decimal (BDT)	No	Latest known unit price, overwritten on update
Delivery charge	Decimal (BDT)	No	Flat charge, overwritten on update, no history kept
Minimum quantity	Decimal	No	Low stock threshold, default 0
Manufacturer	Text	No	
Datasheet URL	URL	No	
Created on / Updated on	Datetime	Auto	System maintained

FR-1.2 The system shall support create, read, update and archive of components. Hard deletion is blocked when stock quantity is greater than zero or movement history exists.

FR-1.3 The component list shall be a server paginated table with the columns: image thumbnail, name, part number, category, unit, total quantity, last buying price, delivery charge, locations, updated on.

FR-1.4 The list shall be sortable on name, part number, category, total quantity, last buying price and updated on.

FR-1.5 The list shall be filterable by category, unit, stock status (in stock, out of stock, below minimum), price range and project usage.

FR-1.6 Free text search shall cover component name, part number, description and location code.

FR-1.7 The system shall provide two Excel/CSV importers:

- Library import: creates or updates component definitions
- Inventory import: creates or updates stock quantities against location codes, referencing components by part number or name

FR-1.8 Each importer shall provide a downloadable template, a column mapping step, and a validation preview showing rows to be created, rows to be updated and rows rejected with the reason. The import shall commit atomically and produce a downloadable error report.

FR-1.9 The inventory importer shall reject any row whose target chamber already holds a different component, and any fractional quantity for a unit that does not allow fractions.

FR-1.10 The system shall export the component list and the inventory list to Excel/CSV, honouring the active filters and the selected columns.

FR-1.11 The component detail page shall show current stock by location, full movement history, requisition history, and projects the component has been consumed against.

1.2 Categories

FR-2.1 A category record shall hold a name (required, unique) and an optional description.

FR-2.2 The system shall support full create, read, update and delete of categories. Deletion is blocked while components are assigned; the user is offered a reassign then delete flow.

FR-2.3 The category list shall be a table with the columns: category name, component count, total stock quantity.

FR-2.4 Selecting a category shall open the component list filtered to that category.

1.3 Units of Measure

FR-3.1 A unit record shall hold a name (for example Pieces), a symbol (for example pcs) and a fractional flag.

FR-3.2 The system shall ship with seeded units: pcs, metres, centimetres, grams, kilograms, rolls, sets. Admin may add, edit and delete units.

FR-3.3 Deletion of a unit is blocked while components reference it.

FR-3.4 Quantities shall be stored as decimal with three decimal places. When a component's unit has the fractional flag disabled, the system shall reject any non integer quantity on every operation and import.

FR-3.5 All quantities shall be displayed with the unit symbol throughout the application and in all exports.

1.4 Requisition Requests

FR-4.1 Any authenticated user shall be able to raise a requisition containing: component reference or free text part name for a part not yet in the library, quantity, unit, project (optional), needed by date (optional), reason or note. Requester and created on are set automatically.

FR-4.2 The requisition status lifecycle shall be as follows, with a single approval step performed by Admin.

Status	Set by	Next allowed status
Pending	System, on creation	Approved, Rejected, Cancelled
Approved	Admin	Ordered, Cancelled
Ordered	Admin	Received, Cancelled
Received	Admin	Terminal
Rejected	Admin	Terminal
Cancelled	Admin or the requester while Pending	Terminal

FR-4.3 Only Admin shall change requisition status. Every change shall record the actor, the timestamp, the previous status and an optional note, and the full history shall be visible on the requisition detail page.

FR-4.4 The requisition list shall be a paginated table with the columns: request ID, component or requested part name, quantity, requested by, project, status, created on, needed by, days in current status. It shall be filterable by status, requester, project and date range.

FR-4.5 On transition to Received the system shall offer a receipt flow that: creates the component in the library if the requisition was raised as free text, adds the received quantity to a chosen chamber, and updates the component's last buying price and delivery charge with the actual amounts paid.

FR-4.6 The requester shall receive an in-app notification on every status change of their requisition.

1.5 Roles and User Management

FR-5.1 The system shall provide three fixed roles: Admin, Manager and Staff. Roles are seeded and are not user definable in version 1.

FR-5.2 Access shall follow the permission matrix below.

Capability	Admin	Manager	Staff
Manage users and roles	Yes	No	No
Configure cabinet and drawers	Yes	No	No
Manage categories and units	Yes	Yes	No
Create and edit components	Yes	Yes	No
Excel import and export	Yes	Yes	No
Add stock	Yes	Yes	No
Take and return stock	Yes	Yes	Yes
Transfer stock	Yes	Yes	No
Adjust stock	Yes	No	No
Raise requisition	Yes	Yes	Yes
Change requisition status	Yes	No	No
Manage projects	Yes	Yes	No
View reports	Yes	Yes	Own activity only
View audit log	Yes	No	No

FR-5.3 Admin shall be able to create users, edit users, assign roles, deactivate users and trigger a password reset. Users are never hard deleted.

FR-5.4 The system shall provide email and password authentication, session handling, logout, password change and password reset by email link.

FR-5.5 Role permissions shall be enforced on the server for every API endpoint, not only hidden in the user interface.

1.6 Component Inventory

FR-6.1 A stock entry shall hold: component reference, chamber location, quantity, description or note, created on, updated on.

FR-6.2 A chamber shall hold at most one component type. The system shall enforce uniqueness of the location across stock entries and reject any operation that would place a second component into an occupied chamber.

FR-6.3 A component may be stocked in more than one chamber. Total quantity of a component is the sum across its chambers.

FR-6.4 The system shall support the following stock operations.

Operation	Effect	Required inputs	Allowed roles
Add	Introduces new stock into a chamber	Component, location, quantity, optional unit price and delivery charge	Admin, Manager
Take	Decreases quantity in a chamber	Location, quantity, optional project, optional note	All roles
Return	Increases quantity in a chamber	Location, quantity, note, optional link to the original take	All roles
Transfer	Moves quantity between chambers	Source location, destination location, quantity	Admin, Manager
Adjust	Corrects quantity after a physical count	Location, new quantity, mandatory reason	Admin

FR-6.5 Every operation shall write an immutable movement record holding: type, component, source location, destination location, quantity, unit price where applicable, user, project, note and timestamp. Movements are never edited or deleted; corrections are made by posting a new movement.

FR-6.6 Take and Transfer shall not exceed the quantity available at the source chamber. Quantity on every operation must be greater than zero.

FR-6.7 Transfer to an occupied chamber shall be permitted only when the destination already holds the same component.

FR-6.8 When a chamber reaches zero quantity the stock entry shall be retained with quantity zero for a configurable retention period, after which the chamber is released as empty.

FR-6.9 Project tagging on a Take operation shall be optional.

FR-6.10 A component whose total quantity is at or below its minimum quantity shall be flagged as low stock in the component list, on the drawer map, on the dashboard and in reports.

FR-6.11 Concurrent stock mutations shall be serialised so that two simultaneous takes cannot drive a chamber below zero.

1.7 Drawer Interface

FR-7.1 Version 1 shall support one cabinet. The cabinet is defined by a row count from 1 to 9 and a column count from A to Z, configurable by Admin.

FR-7.2 Each drawer shall have between 1 and 9 chambers, defaulting to 1. The chamber count is set per drawer and may be changed by Admin.

FR-7.3 A location code shall take the form Column letter + Row number + Chamber number. Row 1, column A, chamber 1 is A11. Codes shall be unique and validated against the current cabinet dimensions.

FR-7.4 Reducing the cabinet dimensions or a drawer's chamber count shall be blocked while any removed chamber holds stock.

FR-7.5 The drawer view shall render the cabinet as a 2D grid. Each drawer cell shall display its code, its chamber subdivisions and an occupancy state.

State	Meaning	Indicator
Empty	No component assigned to the chamber	Neutral / grey
Occupied	Component present, above minimum quantity	Green
Low stock	Component present, at or below minimum quantity	Amber
Out of stock	Component assigned but quantity is zero	Red

FR-7.6 Selecting a drawer shall open a panel listing each chamber, the component it holds and the quantity, with quick actions for Take, Return and Transfer.

FR-7.7 Searching a component from the drawer view shall highlight every chamber holding it.

FR-7.8 The drawer view shall provide a printable label sheet containing the location code and a QR code for each chamber. Scanning a QR code shall open that chamber's panel.

1.8 Projects

FR-8.1 A project record shall hold: name (required, unique), description, status of Active or Closed, created on and created by.

FR-8.2 The system shall support full create, read, update and close of projects. Deletion is blocked once movements or requisitions reference the project.

FR-8.3 Take operations and requisitions may optionally be tagged with a project. Closed projects shall not be selectable on new operations.

FR-8.4 The project detail page shall list components consumed, quantity per component, and estimated value calculated as quantity multiplied by the component's last buying price, together with linked requisitions.

1.9 Reports

FR-9.1 All reports shall support an on screen table view, a date range where applicable, and export to Excel or CSV.

FR-9.2 The system shall provide the following reports.

Report	Contents
Current stock	Component, category, unit, total quantity, chambers held, last buying price, delivery charge, estimated stock value
Low stock	Components at or below minimum quantity, with the shortfall quantity
Movement history	All movements filtered by component, user, project, type, location and date range
Project consumption	Components and quantities consumed per project with estimated cost
Requisition status	Requisitions grouped by status with ageing in days

Report	Contents
Drawer utilisation	Occupied against empty chambers, plus the list of free location codes
User activity	Take and return volume per user over a date range

FR-9.3 The dashboard shall display: total components, total stock value, low stock count, out of stock count, pending requisitions, drawer utilisation percentage, recent movements and the most consumed components.

1.10 Cross Cutting

FR-10.1 The system shall maintain an audit log of every create, update and delete on components, categories, units, users, projects, the cabinet layout, and every requisition status change, recording actor, action, entity, before and after values and timestamp.

FR-10.2 The system shall raise in-app notifications for requisition status changes and low stock events.

FR-10.3 All timestamps shall be stored in UTC and displayed in Asia/Dhaka.

FR-10.4 All monetary values shall be in BDT and displayed with the Taka symbol and two decimal places.

2. Non-Functional Requirements

ID	Requirement
NFR-1	Performance: the component list loads in under 2 seconds for 10,000 components using server side pagination at 25 rows per page. Search returns in under 1 second.
NFR-2	Scale: 10,000 components, 50,000 stock entries, 500,000 movements and 50 concurrent users without degradation.
NFR-3	Security: role checks enforced server side on every endpoint, hashed passwords, CSRF protection, rate limited login, uploaded files validated by type and size and served from a non executable path.
NFR-4	Data integrity: stock mutations execute inside a database transaction with row level locking. The movement ledger is append only.
NFR-5	Availability: nightly database backup with 7 day retention, weekly media backup, documented restore procedure.
NFR-6	Usability: responsive down to tablet width so the drawer map is usable on the shop floor. Keyboard first search on all list screens.
NFR-7	Browser support: the latest two versions of Chrome, Edge, Firefox and Safari.
NFR-8	Observability: application error logging with alerting, and audit log retention of at least 12 months.
NFR-9	Localisation: English interface, Asia/Dhaka timezone, BDT currency.

3. Technology Stack

Layer	Choice	Rationale
Backend	Django + Django REST Framework	Fast admin scaffolding, mature permissions layer
Database	PostgreSQL	Transactional integrity for the movement ledger
Async tasks	Celery + Redis	Excel import and export, report generation, emails
Frontend	Basic HTML + CSS + JS	Basic HTML, so that raspberry pi can handle rendering
File storage	Local volume or S3 compatible bucket	Component images and label sheets
Spreadsheets	openpyxl	Import validation and export generation
Deployment	Docker, nginx, VPS	Single tenant internal deployment

4. Assumptions and Constraints

- The system serves one organisation on a single deployment. There is no tenant isolation requirement.
- There is exactly one physical cabinet in version 1. Location codes are therefore globally unique without a cabinet prefix.
- A chamber holds at most one component type. Mixed storage is not supported.

- Only the latest buying price and a flat delivery charge are stored per component. Historical purchase prices and supplier records are not tracked.
- Requisitions require a single approval by an Admin. There is no multi level approval chain.
- Stock valuation uses the last buying price and is indicative only. It is not an accounting figure.

5. Future Scope

The following are deliberately excluded from version 1 but should be accommodated by the data model where the cost is low.

- Bill of materials: define a BOM per project or board, check stock coverage and auto generate requisitions for shortfalls
- Multiple cabinets, with cabinet prefixed location codes
- Purchase price history with supplier, date and lead time
- Purchase order generation from approved requisitions
- Barcode and QR scanning from a phone camera for take and return
- Distributor catalog lookups to auto fill part data and pricing
- Physical stock take mode with variance reporting
- Scheduled email digests for low stock and pending requisitions

Design note

The movement ledger, the unit reference on each component, and the optional project tag are the three structures that make the future scope items cheap to add later. They should not be simplified out of version 1.